

Python Interview Practice



all possible method and problems set to practice and add must tech giant company top interview questions of python language on basis topics

- basic & syntax
- all datatypes related
- loop related
- condition related
- user input related

Here's a **complete Python interview + practice set** based on your topics, including **top tech company style questions (Google, Amazon, Microsoft, Meta style patterns)**.

I'll organize it so you can practice step-by-step like a roadmap.

1. BASIC & SYNTAX (Fundamentals)

◆ Concepts to Learn

- Variables
- Keywords
- Indentation rules
- Comments
- Type conversion
- Operators

Practice Problems

1. Print "Hello World"
2. Swap two variables (with/without temp variable)
3. Check if a number is positive, negative, or zero
4. Convert Celsius → Fahrenheit
5. Find area of circle
6. Find simple interest
7. Check even or odd
8. Find largest of two numbers
9. Find largest of three numbers

10. Compute power (a^b)

Top Company Interview Questions

- Explain difference between **Python vs C/C++ syntax**
- What are Python keywords?
- Why indentation is important in Python?
- Difference between `is` and `==`
- What is dynamic typing?

2. DATA TYPES (VERY IMPORTANT FOR INTERVIEWS)

◆ Concepts

- int, float, string, bool
- list, tuple, set, dictionary
- type casting
- mutable vs immutable

Practice Problems

1. Convert string $\rightarrow$ int, float
2. Reverse a string
3. Count vowels in string
4. Find frequency of characters
5. Remove duplicates from list
6. Convert list $\rightarrow$ tuple
7. Convert tuple $\rightarrow$ list
8. Find max/min in list
9. Merge two dictionaries
10. Check palindrome string

Top Company Questions

- Difference between list, tuple, set, dictionary
- Why tuple is faster than list?
- Why set cannot store duplicates?

- What is hashing in dictionary?
- What happens internally in Python dict?

3. LOOP RELATED (VERY HIGH FREQUENCY IN INTERVIEWS)

◆ Concepts

- for loop
- while loop
- nested loops
- break, continue, pass

Practice Problems

1. Print 1 to N numbers
2. Print even numbers in range
3. Print multiplication table
4. Sum of first N numbers
5. Factorial of number
6. Fibonacci series
7. Reverse a number
8. Count digits of number
9. Pattern printing:
 - *
 - **
 - _____
10. Prime number check

Top Company Questions

- Difference between `for` and `while`
- When to use `break` vs `continue`?
- How does Python loop internally work?
- Time complexity of loops
- Can Python loop over dictionary?

4. CONDITIONAL STATEMENTS (LOGIC BUILDING CORE)

◆ Concepts

- if, elif, else
- nested conditions
- logical operators (and, or, not)

Practice Problems

1. Check leap year
2. Check voter eligibility
3. Grade system (A, B, C)
4. Find max of 3 numbers
5. Check triangle validity
6. Check Armstrong number
7. Check palindrome number
8. Check prime number
9. Calculator using conditions
10. Even/odd without modulo

Top Company Questions

- Difference between `elif` and multiple `if`
- What is short-circuit evaluation?
- Can if condition return values?
- Boolean evaluation rules in Python
- Truthy vs falsy values

5. USER INPUT RELATED (VERY IMPORTANT FOR CODING ROUNDS)

◆ Concepts

- `input()`
- type conversion

- string input handling
- multiple inputs

Practice Problems

1. Take name and greet user
2. Input two numbers and add
3. Input list of numbers
4. Find average of inputs
5. Input sentence and count words
6. Input multiple values in one line
7. Find largest input number
8. Input string and reverse it
9. Input number and check prime
10. Input marks and calculate grade

Top Company Questions

- Why input() returns string?
- How to take multiple inputs in Python?
- Difference between input() and sys.stdin
- How to optimize input for large data?
- What happens if invalid input is given?

BONUS: TOP TECH GIANT CODING PATTERNS

These are **very common in Google / Amazon / Microsoft interviews**:

Must Practice Problems

- Two Sum (Array logic)
- Reverse String
- Valid Parentheses
- FizzBuzz
- Find missing number
- Maximum subarray sum
- Frequency counter problems

- Sorting-based logic problems
 - Remove duplicates efficiently
 - Sliding window basics
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HOW TO PRACTICE (IMPORTANT)

Use this method:

Step 1:

Learn concept

Step 2:

Solve 5–10 problems

Step 3:

Re-solve without seeing solution

Step 4:

Explain out loud (interview simulation)

If you want next step:

I can also give you:

-  Daily 30-day Python interview roadmap
-  MCQ quiz for each topic
-  Google/Amazon coding questions with solutions
-  LeetCode beginner → advanced path

Just tell me 